Homework 5: Linear Transformations

BCS A, E, F, G

Problem 1. ***Finding Standard Matrix of the Derivative.***

a. Let $B = \{e^x, xe^x, x^2e^x\}$ be a basis of the vector space $V = span(B)$. Find the standard matrix $[D]_B$ of the derivative map $D: V \to V$ with respect to B.

b. Use this standard matrix to find fourth derivative of $x^2e^x + 2xe^x$.

c. Use this standard matrix to compute the following integral.

$$\int (3x^2e^x - xe^x)\, dx\ .$$

Problem 2. ***Finding Standard Matrix of the Derivative..***

Let $B = \{\sin x, \cos x\}$ and $V = span(B)$. Find a square root $D^{1/2}$ of the derivative map $D: V \to V$ and use it to compute $D^{1/2}[\sin x]$ and $D^{1/2}[\cos x]$.

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Good Luck